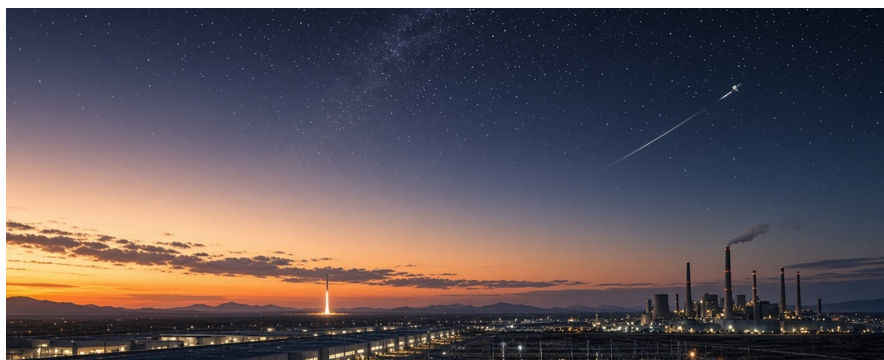




Data Centers 2026

Executive summary — land, iron, heat. Orbit is still a filing.

This is a working synthesis drawn from the illustrated global briefing dated 19 September 2026, a collaboration between **Roy Poystila** and **Grok (xAI)**. It is not an offering document, not an audit, and not a forecast to trade on. Announced gigawatts are not live megawatts. Where a sentence is a public filing or a published rack rating, treat it as reportage. Where it is about what a player is optimizing for, treat it as a map, not a verdict.

The full briefing runs fourteen sections plus a glossary. This file keeps the argument, the binding constraints, the named projects, and the distinctions that stop a slide from being mistaken for a feeder. It drops the plate-by-plate tour.

The sentence to keep

A data center is a building that keeps other people's computers running without pause: power, cooling, network, physical security, and someone awake if a pump fails. Almost nothing treated as “the internet,” “the cloud,” or artificial intelligence (AI) lives on a laptop. It lives in one of these rooms. Banks, hospitals, retailers, governments, search, maps, and streaming all rent time in the same industrial object.

The AI wave did not invent the building. It split the product. **Training** a frontier model is a weeks-long industrial run that wants cheap firm power and does not care about twenty milliseconds. **Inference** — answering a person now — wants to sit near users and data, on dense fiber. The industry now sells a factory in a field and a hall on a metro ring. They look like one business. They are not one siting problem.

Directories count on the order of 11,000–12,000 data centers on Earth. Most barely move the power total. Global information-technology (IT) load is estimated near **161 gigawatts in 2026**, up from about 123 last year. Hyperscale tallies are the better object: roughly 1,300 operating hyperscale sites and about 770 in the announced pipeline. Analysts talk about ~100 GW of new installed capacity this decade. Compounding demand cases wave at ~491 GW by 2030. Those two numbers cannot both be operational facts. The gap is turbines, transformers, hearings, and water.

The industry is already on two tracks. Track one is pouring next to gas plants and existing substations — Meta in Louisiana and Alberta, Amazon in Indiana, Microsoft pairing halls with a nuclear restart and West Texas gas, xAI filling industrial shells in Memphis. Track two is a press release waiting on a permit, a queue, or a piece of electrical iron that takes three to four years to manufacture. A campus beside a plant still sits dark if the transformer was not ordered years ago.

The working rule

Power and electrical iron decide where a hall can live. The chip rack decides what the hall must be. Politics decides whether it opens. Fiber, export licenses, and insurance decide which permitted sites are usable. Everything else — nuclear contracts, custom chips, underwater pods, satellites — is an attempt to escape someone else's queue.

1. What the buildings are for

Five buckets still organize the directory. They now describe different physics, not just different leases.

Type	Who	What it now requires
Hyperscale / AI factory	AWS, Azure, Google, Meta, Oracle, xAI	100 MW to multi-GW campuses; liquid cooling; often own generation.
Wholesale / retail colo	Equinix, Digital Realty, Vantage, QTS	Landlord sells a powered shell. Time-to-power is the product.
Enterprise	Banks, governments, corporates	Still most of site count. Losing the new-gigawatt race.
Edge / micro	Telco, CDN, metro cache	Latency. Not where frontier training lives.
Sovereign / HPC factory	National labs, Gulf funds, EuroHPC	Same liquid halls, different owner, export rules, mandate.

Counting sites and counting megawatts now tell opposite stories. Enterprise still wins the headcount of rooms. New gigawatts are going to factories specified backward from a liquid rack. Colocation remains swing capacity for whoever cannot wait on their own plant. The migration that matters for 2026–30 is not enterprise-to-cloud. It is cloud-to-factory.

Training can leave town. Inference mostly cannot.

Treat “AI load” as two jobs or the map lies. Training can sit next to a gas plant in the prairie. Inference is bound by data gravity: the model wants the user, the data, and a peering hotel in Ashburn, Dallas, Dublin, Frankfurt, or Tokyo. A 200-millisecond trip to Alberta is fine at the end of a two-week training run. It is a product failure for a copilot. Off-grid gas and nuclear co-location are therefore a *training-first* escape. Northern Virginia and the European FLAP-D core — Frankfurt, London, Amsterdam, Paris, Dublin — do not empty out because Texas and Alberta are pouring.

Two filters sit above cheap power. Inference needs the right cable, not only a nearby city. And United States export rules, end-use checks, and data-residency law decide whether a frontier cluster may legally exist on a given soil. Gulf and European “sovereign” campuses are license-and-jurisdiction buildings wearing a cooling plant.

2. Markets, iron, and what the queue hides

Northern Virginia remains the largest interconnection market on Earth and the cautionary tale: multi-gigawatt inventory on a grid that is now a political object. Prince William’s Digital Gateway fight taught operators that a feeder has voters. Texas is the sprint because land is cheaper and Electric Reliability Council of Texas (ERCOT) interconnects were faster than PJM Interconnection — which is why the state is now auditing a tens-of-gigawatt queue and why towns such as San Marcos have reached for local bans. FLAP-D is tight on power and water at once. The Nordics sell warm-water heat into a city network. The Gulf is buying sovereign campuses into the worst ambient-heat climate on the map.

Market	Role	Binding bottleneck
Northern Virginia	Interconnection capital	Transmission; household rates; land use
Texas (ERCOT)	AI mega-campus sprint	Queue audits; municipal bans; summer derating
FLAP-D	European core	Grid moratoria; power and water permits
Nordics / Helsinki	Heat-reuse hubs	Fiber egress; smaller local markets
Gulf (Abu Dhabi / UAE)	Sovereign AI bets	Ambient heat; water; geopolitics
Singapore / Tokyo	APAC interconnection	Power caps, land, tight quota

Substation iron is now a co-equal bottleneck with turbines

Gas-turbine order books sold out into the early 2030s are widely tracked. Less tracked: high-voltage generator-step-up (GSU) transformers and medium-voltage switchgear with manufacturing lead times of three to four years. Siting a hall next to a wellhead or a restarted reactor solves fuel. Without a transformer ordered in 2023 or 2024, the hall is a warehouse. “Behind-the-meter” is a procurement calendar that starts before the groundbreaking photo.

The interconnection list is not the whole risk file. Flood, wildfire, hail, and extreme-heat days are now insurance and lender questions. Chillers, coolant-distribution units, busbars, generators, batteries, fire-suppression a marshal will sign for a liquid hall, and copper all slip on their own clocks. Labor still clusters in the old metros. Software getting cheaper per token can make some announced gigawatts optional. Fifteen-year paper against a three-year rack is a different asset at 3 percent than at 7. Cheap renewable certificates are not power on the feeder at 4 p.m. in July.

3. Who is spending, and what is actually being poured

2026 capital-expenditure guidance across the top operators sits in a **\$700–850 billion** band depending on who is counted. Spend is outrunning operating cash flow at several firms. Treat every figure as guidance, not audited concrete.

Operator	2026 guidance	Pointed at
Amazon / AWS	~\$200B+	Rainier (Indiana); Trainium; owned campuses plus leased regions.
Microsoft	~\$175–195B	Fairwater halls; OpenAI; Crane / TMI nuclear; West Texas gas with Chevron.
Alphabet (Google)	~\$180–190B	TPU v6/v7; Visakhapatnam with Adani; on-site turbines after a late start on gas.
Meta	~\$125–145B	Hyperion (Louisiana); Prometheus; El Paso; Sturgeon County; MTIA.
Oracle	~\$65B class	Stargate / Abilene; 50,000-GPU AMD Helios as dual-source proof.
xAI / SpaceX compute	cluster + filings	Colossus Memphis; Starmind FCC file; Terafab framework.

Named objects, not slide titles

Meta Hyperion (Richland Parish, Louisiana) is the template: multi-gigawatt class, talked near 5 GW over phases, next to on-site gas so the campus does not wait out a transmission queue. Prometheus, El Paso, and Sturgeon are the same idea in other dirt. The nuclear power-purchase-agreement book is the 2030s story, not 2026 energization.

AWS Rainier (New Carlisle, Indiana) is the Anthropic-weighted training campus and a test of whether a Midwestern grid node can absorb factory-scale load without becoming Northern Virginia. Amazon owns the hall for frontier training and still leases colo for regional inference. Trainium is how AWS tries not to pay list price on every token.

Microsoft Fairwater plus **Crane / Three Mile Island Unit 1** are a pair. Fairwater is the liquid hall (Mount Pleasant, Wisconsin, came online in 2026 and immediately collected a noise complaint). Crane is 835 megawatts for twenty years, restart talked for 2028 rather than the earlier 2027 slide. The Chevron West Texas behind-the-meter gas plant (~2.7 GW class in public descriptions) covers the years before those nuclear electrons exist. Microsoft sequenced power. It did not solve it.

Stargate is two bets that share a name: Texas speed on gas and interconnect muscle (Oracle, OpenAI, SoftBank, Crusoe), and Abu Dhabi with G42 as sovereign capital buying a training map that does not have to ask PJM. First Texas clusters have been talked for late 2026. Full announced gigawatts sit well beyond first power. Early-2026 U.S. tracking already had on the order of 75 projects blocked or delayed in a single quarter.

Phantom load

Utilities have a name for queue inflation: phantom load. Capgemini's June 2026 survey of more than 600 electricity executives (*AI meets the grid*) found that 67 percent see speculative data-center requests, and that about 19 percent of *those* flagged requests never become live megawatts. That is not one in five watts on Earth. Developers also shop the same server allocation across several utilities at once. When one twin withdraws, the remaining queue often does not simply move up — the study was done for a specific load at a specific bus. A restudy can take six months to more than a year. A 21.5 GW Alberta interest queue or a 50 GW-class ERCOT book cannot be read as future construction.

4. Canada — buildings in the east, megawatts in Alberta

Canada is the cleanest split-screen in the briefing. Live stock sits in Ontario (about 110 sites), Quebec, and British Columbia — the old hydro-and-fiber provinces. Something like nine-tenths of the proposed megawatt pipeline has filed in Alberta. One public split: on the order of 21.5 GW in the filed interest queue against about 9.8 GW announced, versus about 1.75 GW announced in Ontario.

The Alberta Electric System Operator 1,200 MW interim large-load allocation is already spoken for: Meta's Sturgeon County campus at 970 MW and Keephills Phase I at 230 MW. Meta's roughly \$13 billion Sturgeon project is paired with the adjacent ~\$4.6 billion, 932 MW Greenlight gas plant. That pairing is the Canadian sentence: you may have a grid allocation, and you still bring your own plant. Alberta's bargain is bring-your-own-power, a 120-day shot clock, and an equipment levy so household rates do not underwrite compute. Feeding on the order of 6 GW of planned on-site generation implies billions in new gas-pipeline capacity. Quebec's Bill 69 (legislative sign-off above 5 MW) and B.C. large-load caps mean hydro provinces are structurally harder to enter than gas Alberta, because clean power is already promised to decarbonizing industry. Canada is not short of electrons in the abstract. It is short of electrons allowed to become a data-center feeder this decade.

5. SpaceX — three stacked stories, one of them live

Headlines treat SpaceX compute as one story. It is three. Collapse them and you award orbit a gigawatt it has not flown.

Ground — Colossus. xAI's Memphis clusters are the terrestrial speed run: industrial shells converted in months, tens of thousands of graphics processing units (GPUs), gas on the fence. This is the only layer already renting compute at scale. Google renting time on that capacity is the tell that even rivals will buy a messy cluster if the alternative is a queue.

Orbit — Starmind. January 2026 FCC application SAT-LOA-20260108-00016 for an orbital data-center system of up to one million satellites. First-generation AI1: ~20 m tall, 70 m deployed wingspan, ~4 tonnes, ~120 kW average / 150 kW peak compute, ~110 m² liquid radiators. Results home over Starlink lasers. Prototypes talked for late 2027. Company remarks have put 100 GW in orbit at "on the order of 10,000 Starship launches." SpaceX's own S-1 is colder: the plan rests on unproven technologies, or technologies that do not exist.

Silicon — Terafab. The attempt to stop buying the next increment from the same Taiwan Semiconductor Manufacturing Company (TSMC) / NVIDIA queue. Grimes County, Texas, talked as on the order of 100 million square feet across phases. Intel is the process partner — 18A in some early descriptions, 14A as the full-scale node, with the 14A process-design kit not expected until late 2026. Construction start listed around 1 December 2026; first-phase completion spoken in the 2028 window. Headline dollars run from the mid-tens of billions to a \$119 billion four-phase envelope. The S-1 still calls Terafab a general framework without binding commitments that Tesla or Intel remain indefinitely. Without radiation-tolerant logic, memory, and packaging in-house, an orbital fleet is another purchase order against TSMC's Chip-on-Wafer-on-Substrate packaging line.

Proven versus planned

Through mid-September 2026 Starship was not a bus route. Only two flights had flown in 2026 before Flight 14. Flight 13 (24 July) was another suborbital splashdown. Flight 14 — slipped toward late September — was the intended first orbital mission: ~275 km, six orbits, first live payload of roughly 20 Starlink V3 satellites. Neither stage was scheduled to catch. Pad 1 at Starbase is in a deep refit into 2027. An AI1 craft does not fly on Falcon 9. V3 already does not.

Case	Launches / year	Implied daily rate
Optimistic 5-year replenishment of a million-sat class system	~3,500 Starship flights	~10 / day
Pessimistic pack-out, or 100 GW-class company remarks	~10,000+ flights	~27 / day
2025 world record, every rocket on Earth	329 orbital attempts	~0.9 / day

Even the friendly case is a twenty-fold jump in SpaceX launch rate. Last year's world record was 329 attempts, SpaceX flying about half. Heat is the other half. AI1 wants roughly twice the International Space Station heat load through about a quarter of the radiator area. Starcloud's flown H100 remains the only public GPU-in-orbit data point, and it could not run at full power because the radiator was too small. Radiation shielding fights the FCC's 15-joule casualty threshold on re-entry. Until Flight 14 and an AI1 prototype exist as data, treat orbital gigawatts as a filing, not a supply plan.

6. Who writes the hall, who fills the map, who writes the power contract

NVIDIA — the drawing most halls are poured around

Land and power decide where a hall can exist. The rack decides what the hall is. NVIDIA still writes that drawing. Jensen Huang has cited on the order of a trillion dollars of Blackwell and Rubin system visibility through 2027 — a demand signal, not a delivery schedule. Hopper-era air halls lived near 40 kW per rack. The GB200 NVL72 pushed the industry to ~120–130 kW liquid. Vera Rubin VR200 is specified around 190–230 kW, fanless, direct-to-chip, ~45°C inlet. Kyber is drawn near 600 kW for 2027. A one-megawatt-class Feynman rack is the next slide. Treat Feynman as a drawing in 2026, not a stock-keeping unit.

NVIDIA now sells a rack that behaves like one accelerator, the fabric that ties racks into a hall (NVLink inside, Spectrum-X / InfiniBand outside), the software that keeps models there (CUDA), and — as of August 2026 — a way for someone else to finance the factory. Partnerships with Apollo, BlackRock, Blackstone, Brookfield, Goldman Sachs and KKR were framed as platforms meant to mobilize on the order of \$500 billion of third-party capital. That is company and analyst language, not cash already deployed. The intent is the point: NVIDIA wants the factory to be an asset class.

The financing mismatch does not show up on a cooling diagram. Project-finance and data-center debt is still often written on a 15- to 20-year amortization. NVIDIA's public cadence is closer to 12 to 24 months per generation. A campus poured for 120 kW liquid can need a retrofit long before the loan is a third of the way through. If the hall cannot be replumbed, the concrete is a husk and the credit is long. NVIDIA does not own the feeder and does not rezoning-fight Prince William. It writes the drawing those fights are about.

Amazon — the company that fills the map

Amazon Web Services is still the largest cloud. In 2026 it is also the largest single capex book in the sector; mid-year calls moved guidance from about \$200 billion toward \$220 billion. AWS talks a backlog of customer commitments on the order of half a trillion dollars. Project Rainier is the named AI factory, Trainium-heavy, Anthropic as the anchor — a commitment talked as high as five gigawatts of Amazon AI chips over time. Amazon is not trying to kill NVIDIA this decade. Andy Jassy has said so. It is trying to stop paying list price on every token AWS serves. Trainium2 is in volume; Trainium3 has been talked as ramping through 2026 on TSMC N3. Landscape if this holds: more secondary metros become real AWS regions; Trainium sets a second floor under inference price; NVIDIA remains the guest GPU in the same buildings.

Microsoft — the company that writes the power contract

Microsoft houses a large share of OpenAI's training demand. Calendar-2026 capex has been talked in the \$175–195 billion band; one print put remaining performance obligations at \$678 billion. Cloud revenue still more nearly covers the build than at some peers. That does not make the physics easier. Maia is the owned-silicon hedge on Azure traffic. It does not replace the NVIDIA halls OpenAI trains on. Landscape if this holds: off-grid gas then a nuclear PPA becomes normal for a frontier campus; community noise and rate fights travel with every Fairwater-class site. Landscape if it slips: Crane slips past 2028, West Texas gas is the whole decade, and Microsoft is a very large NVIDIA customer with a good treasurer.

The rest of the field

The anticipated plan among AMD, UALink, and the custom-chip camp is not to kill NVIDIA in 2027. It is to make “NVIDIA or equivalent” a sentence a chief financial officer can say. Oracle’s 50,000-GPU MI450 Helios cluster is the proof point. Google’s Tensor Processing Unit v7 means frontier training can live off CUDA. Meta’s MTIA is how Meta tries not to pay rack-tax on Llama serving. Colo landlords are selling time-to-power before hyperscalers walk around them with a plant. Sovereigns in the Gulf, France, and India are trying to make sure the next decade’s training runs can happen on their soil — which only works if power, cooling, and talent arrive on the same calendar as the chips.

If anticipated plans hold	If they slip
Largest clusters sit next to private generation; grid is for people	Queues and transformer waits cap new GW; campuses stay slides
Halls designed from Kyber / Feynman backward	120 kW liquid halls take write-downs
Inference mix shifts toward ASICs and AMD	NVIDIA keeps serving-price power
A second map for large training (Gulf, sovereign)	Those sites stay inference and fine-tune hubs
Starmind and Terafab are late, small slices of 2030 supply	Orbital GW stays a filing through the decade

7. Heat, capability, and the supercomputing inheritance

A data center used to be judged by square feet and power-usage effectiveness. An AI factory is judged by what a megawatt can do — tokens served, models trained, simulations finished — and by whether the leftover heat is a waste stream or a product. Peak floating-point operations per second still appear on slides. The number starting to travel into utility and finance meetings is tokens per megawatt under a service-level objective. Capability is also becoming time-varying: Silicon Valley Power has already sent hundreds of demand signals to an AI factory that dropped load while keeping priority jobs alive.

Air cooling did not fail. The rack outgrew it. You cannot blow 200–600 kW out of a box with computer-room air handlers at a temperature or noise a human hall will accept. National labs learned this on Titan, Summit, Frontier and their European twins: capture heat at the cold plate, move it in water, reject it at the plant or sell it. NVIDIA’s Rubin generation is specified at up to 45°C inlet — hotter than a hot tub — and fanless. Warm coolant lets dry coolers work most of the year and can drop facility water use from on the order of 2.6 million gallons per megawatt-year on cooling-tower plants toward zero *in the right climate*.

Warm water is not free in a desert

Dry coolers lose approach temperature when the outside air is already hot. In Texas, the Gulf, and the Middle East an operator either derates compute on the worst afternoons or turns evaporative trim back on — which is the water story the 45°C spec was supposed to end. Nordic heat-reuse economics do not transfer to Abu Dhabi. A chiller-less claim is a climate claim.

Once the loop exits at 45–65°C, the heat is finally hot enough to use. LUMI already feeds Kajaani’s district heating. TU Dresden’s high-performance computing heat pumps lift server water from ~55°C to ~90°C — about 3.9 MW thermal, on the order of 3,700 households. OnZero / Helen in Helsinki is the AI-scale version: ~60 MW thermal, enough in the partners’ telling for roughly 70,000 apartments. EU Energy Efficiency Directive Article 26(6) now makes a reuse assessment mandatory above 1 MW. A campus next to a fourth-generation district-heat network is a different economic object than a campus that can only reject to dry coolers or towers.

8. Pushback — the constraint that is not a turbine

The buildout has a politics, and it is not a cartoon of “not in my backyard” versus progress. Project-delay trackers put on the order of 75 U.S. projects blocked or delayed in a single early-2026 quarter. Local moratoria number in the hundreds depending on how a pause is counted. Polling moved from abstract support for “AI infrastructure” to local resistance once a site has a street address. Four drivers keep showing up.

Household rates. In PJM and Northern Virginia the argument that lands is the claim that transmission upgrades and capacity charges leak onto residential bills. Once a rate rider has a number, a rezoning is a utility hearing wearing a land-use costume. **Water.** Evaporative towers still sit on a large share of the live fleet. Opposition is not ignorance of the 45°C spec; it is disbelief that the spec will hold on the hottest week. **Land use, noise, and hours.** A liquid hall is quieter than a 2018 air hall. It is still a 24-hour industrial object. Fairwater collected that complaint on day one. **Tax abatements.** Headcount per megawatt is low. When the abatement is public and the employment number is small, the political bargain collapses even if the engineering is clean.

Place	Kind of pushback	Why it travels
Prince William, VA	Court + corridor collapse	Heroic rezonings are now treated as fragile.
NoVA / PJM	Rates and riders	Household bills became a siting constraint.
Texas / ERCOT	Queue audit + local bans	Fast interconnects attract their own politics.
New York	Environmental pause	Permission can be slowed without a ban.
Pennsylvania	Bring-your-own-power order	The campus may come if it brings the plant.
Scotland	Parliamentary pause (80–26, 16 Sept 2026)	Hyperscale treated as a national class of use.
Alberta	Cap + BYOP + levy	Megawatts proceed only on provincial terms.
Quebec / B.C.	Legislative / allocation caps	Hydro is not spare power.

Pushback does not stop the industry. It prices and relocates it. Load moves toward gas plants, secondary metros, bring-your-own-power provinces, and — in the filings — toward barges, mines, and orbit. A county that says yes still cannot energize a hall whose iron was not ordered years ago. A desert county that says yes still has a water fight on the first 110°F week.

9. Alternate siting — doors that are not next year’s fleet

When a county says no, or a queue says wait, the industry looks for a door that is not on the parcel. Most of those doors are real as research. Almost none of them are how a model gets trained next year. Behind-the-meter gas is not an alternate. It is the main escape already running.

Approach	2026 status	Unresolved limit
Land hyperscale	Multi-GW pouring where power exists	Queues, hearings, transformers
Behind-the-meter gas	Tens of GW of equipment ordered	Turbines sold out; emissions; gas pipes
Nuclear PPA / SMR	PPAs signed; first electrons ~2027–28	Most of the book is 2030s hardware
Floating barge / SMR	10–100 MW designs; shipyard interest	Maritime rules, weather, scale
Underwater pods	Natick research; China ~24 MW class site	Service access; still needs a cable
Mines / underground	MW niches (Lefdal, Intacture)	Expansion and access
Arctic / free cooling	Narvik-class and Nordic builds	Fiber, talent, winter logistics
Orbit	kW-class demo; million-sat FCC file	Launch rate, heat flux, demise, shells

The announced U.S. hyperscaler nuclear book is on the order of 10 GW; Crane is the first named electrons. Microsoft Project Natick proved the thermal idea of a sealed pier (power-usage effectiveness near 1.07) and did not prove a service model. A 10–100 MW barge is a regional inference node, not Hyperion. Lefdal Mine and Intacture prove rock cooling and a hall that does not dominate a skyline; you cannot pour 2 GW into a worked-out drift. Every one of these is a reaction to a specific terrestrial no. 2027 training runs happen in a warehouse next to generation, with a transformer that was ordered years ago.

10. Bottom line

Two tracks. Four constraints. Three refinements that keep the map from lying.

Track one is live: halls next to existing substations, dedicated gas plants, or already-approved feeders. **Track two** is a press release waiting on a queue, a transformer, an environmental review, or a county vote.

Four sentences that should survive the next earnings season. Power — including the unglamorous iron of transformers and switchgear — is the constraint; land and capital are easier. The rack writes the building, and the building's useful life is shrinking as density doubles. Private generation is how serious players leave the household feeder. Orbital, underwater, and barge schemes are real research and real filings; they are not the 2027 training fleet.

Every serious player is trying to escape someone else's queue — GPUs, turbines, transformers, interconnection, or the county. Gas and NVIDIA racks are the escapes already running. Nuclear, UALink, Starwind, and Terafab are the escapes that will decide whether 2030 looks like a thicker version of 2026 or a different industry.

Three refinements. Off-grid gas and remote prairie acreage solve frontier *training*; they do not retire inference in Ashburn or Frankfurt. Interconnection queues are not a census: Capgemini's 2026 survey put phantom or speculative requests in front of two-thirds of electricity executives, with about 19 percent of those flagged requests never becoming load, and duplicate filings trigger restudies that stall the projects that are real. And the hall-as-asset story has a tenor problem — 15- to 20-year debt against a rack cycle that can force a replumb in three to five. A site can have power and still fail an export license, a residency rule, or an insurance file. Software getting cheaper per token can make some slides optional. Some of the announced gigawatts were never even a single building. They were the same building filed in three queues.

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